# 1. Project Title

## Basic Details of Indian School Data

Basic Details of  School Dataset — 500 Records | 25+ States

*Data Visualization & Analysis using Microsoft Excel and Power BI*

| Field | Details |
|---|---|
| Project Name | India School Data — UDISE Analysis & Visualization |
| Domain | Education \| Public Sector Analytics |
| Data Source | Unified District Information System for Education (UDISE), Ministry of Education, Government of India |
| Tools Used | Microsoft Excel 365, Power BI Desktop, Python (openpyxl) |
| Dataset Size | 500 school records \| 59 columns (44 raw + 15 formula) |
| Geographic Scope | Pan-India — 25+ states and union territories |
| Reporting Period | Academic year data (latest available collection) |
| Output Deliverables | Excel Workbook, Power BI Dashboard, Project Documentation |
| Link | Basic%20Details%20of%20Schools_Sample_Data.csv |

# 2. Project Statement

India's school education system is one of the largest in the world, with over 1.5 million schools serving hundreds of millions of children. Understanding the distribution, infrastructure quality, enrollment patterns, and management structures of these schools is critical for policymakers, educators, and researchers to make evidence-based decisions on resource allocation and educational reform.

The UDISE (Unified District Information System for Education) dataset provides granular school-level information collected annually from every recognized school in India — government, private, and aided. This project leverages a representative sample of 500 school records spanning 25+ states to perform structured data analysis and visual exploration.

## 2.1 Problem Statement

> **Core Question:** How are schools distributed across India in terms of category, management, infrastructure, and student enrollment — and what patterns and disparities can be identified?

Key questions this project addresses:

- Which states have the highest school density and student enrollment in the sample?
- What is the split between government-run and private schools, and how does it affect infrastructure?
- How do student-teacher ratios vary across rural vs. urban schools and different management types?
- What enrollment dropout patterns exist across class levels (Primary to Higher Secondary)?
- What trends are visible in school establishment over time, and what do they predict for the future?
- Where are under-resourced schools concentrated, and what prescriptive actions can be recommended?

## 2.2 Objectives

1. Clean, structure, and validate raw UDISE data into analysis-ready format
2. Engineer 15 derived formula columns (ratios, percentages, groupings) for richer analysis
3. Build a 4-sheet Excel workbook with automated summary metrics
4. Develop an interactive Power BI dashboard with 10+ visuals, slicers, and drill-throughs
5. Perform diagnostic, descriptive, predictive, and prescriptive analysis of the dataset
6. Document all findings, methods, and recommendations in this report

## 2.3 Significance

Education data analytics enables targeted interventions — directing teacher recruitment to high-STR schools, identifying regions with low upper-primary retention, pinpointing districts needing infrastructure investment, and forecasting enrollment needs. This project demonstrates how structured data analysis on UDISE data can generate actionable insights even from a 500-record sample.

# 3. Data Description

The dataset is sourced from UDISE, the Government of India's centralized education database. Each row in the dataset represents a single school with its full administrative, geographic, infrastructure, and enrollment profile.

| Property | Value |
|---|---|
| Dataset Name | UDISE School Dataset — India |
| Total Records | 500 schools (rows) |
| Total Attributes | 59 columns (44 source + 15 formula-derived) |
| Geographic Coverage | 25+ Indian states including UP, Rajasthan, MP, Bihar, Karnataka, Maharashtra, AP, West Bengal, Jharkhand, Assam, Odisha, Telangana, Gujarat, Tamil Nadu, Kerala, Punjab, Haryana, |

| Property | Value |
|---|---|
| | Uttarakhand, J&K, Tripura, Mizoram, Meghalaya, Delhi, Himachal Pradesh, Chhattisgarh |
| School Categories | Primary, Primary with Upper Primary, Upper Primary only, Secondary, Secondary with Higher Secondary, and combinations thereof |
| Management Types | 9 types: Department of Education, Local body, Government Aided, Private Unaided (Recognized), Unrecognized, Tribal Welfare Department, Central Tibetan School, Madarsa recognized, Madarsa unrecognized |
| School Types | Co-Educational, Boys Only, Girls Only |
| Location Types | Rural and Urban |
| Establishment Years | 1872 to 2019 (147-year span) |
| Affiliation Boards | State Board, CBSE, ICSE, Others, Blank (primary-only) |

### 3.1 Data Coverage by State (Sample Distribution)

| State | Approx. % of Sample | Dominant School Type |
|---|---|---|
| Uttar Pradesh | ~18% | Primary & Upper Primary (Govt) |
| Rajasthan | ~12% | All categories (Govt & Private) |
| Madhya Pradesh | ~11% | Primary (Govt & Tribal Welfare) |
| Bihar | ~9% | Primary (Dept of Education) |
| West Bengal | ~7% | Primary & Secondary (Govt & Aided) |
| Karnataka | ~6% | All categories (Govt & Private) |
| Jharkhand | ~5% | Primary (Govt & Aided) |
| Maharashtra | ~5% | Primary & Secondary (ZP & Aided) |
| Andhra Pradesh | ~4% | Primary (Local body) |
| Assam | ~4% | Primary (Dept of Education) |
| Other 15+ States | ~19% | Mixed |

## 4. Data Column Details

The dataset contains 59 columns organised into two groups: 44 source data columns (raw UDISE fields) and 15 formula-derived calculation columns added during processing.

### 4.1 Source Data Columns (A – AR)

| Col | Column Name | Data Type | Description / Notes |
|---|---|---|---|
| A | State | Text | Indian state name (e.g., Uttar Pradesh, Karnataka) |
| B | State Code | Integer | UDISE 2-digit numeric state identifier |
| C | District | Text | District name within the state |
| D | District Code | Integer | UDISE district numeric code (blank for some states) |
| E | Sub-District | Text | Tehsil / taluka / sub-district name |
| F | Sub-District Code | Integer | UDISE sub-district code |
| G | Cluster | Text | Administrative school cluster name |
| H | Village | Text | Village or urban locality where school is located |
| I | UDISE Village Code | Text | Unique UDISE village-level identifier |
| J | Pincode | Text | Indian postal PIN code (6 digits) |
| K | Ward | Text | Ward name — populated only for urban schools |
| L | School Name | Text | Official registered school name |
| M | School Code | Text | Unique 11-digit UDISE school identifier |
| N | School Category | Text | E.g., 'Primary', 'Secondary with Higher Secondary' |
| O | School Type | Text | Co. Educational \| Boys Only \| Girls Only |
| P | Management | Text | Governing body type (9 categories) |
| Q | Year of Establishment | Integer | 4-digit year school was founded |
| R | Longitude | Decimal | WGS84 geographic longitude (approx. village centroid) |
| S | Latitude | Decimal | WGS84 geographic latitude (approx. village centroid) |
| T | Status | Text | Operational or Non-operational |
| U | Location Type | Text | Rural or Urban classification |
| V | Class From | Integer | Lowest class taught (1 = Class I, 11 = Class XI) |
| W | Class To | Integer | Highest class taught (5 = Class V, 12 = Class XII) |
| X | Affiliation Board (Secondary) | Text | Board for IX–X: State Board, CBSE, ICSE, Others; blank for primary-only |
| Y | Affiliation Board (H.Sec.) | Text | Board for XI–XII: same codes; blank if school does not go to XII |
| Z | Pre-Primary Rooms | Integer | Number of nursery/kindergarten classrooms |
| AA | Class Rooms | Integer | Standard teaching classrooms |
| AB | Other Rooms | Integer | Labs, library, store rooms, office rooms |
| AC | Teachers | Integer | Total sanctioned and working teacher count |
| AD | Pre-Primary Students | Integer | Students in nursery/pre-primary section |

| Col | Column Name | Data Type | Description / Notes |
|---|---|---|---|
| AE | Class I Students | Integer | Enrolled students in Class I |
| AF | Class II Students | Integer | Enrolled students in Class II |
| AG | Class III Students | Integer | Enrolled students in Class III |
| AH | Class IV Students | Integer | Enrolled students in Class IV |
| AI | Class V Students | Integer | Enrolled students in Class V |
| AJ | Class VI Students | Integer | Enrolled students in Class VI |
| AK | Class VII Students | Integer | Enrolled students in Class VII |
| AL | Class VIII Students | Integer | Enrolled students in Class VIII |
| AM | Class IX Students | Integer | Enrolled students in Class IX |
| AN | Class X Students | Integer | Enrolled students in Class X |
| AO | Class XI Students | Integer | Enrolled students in Class XI |
| AP | Class XII Students | Integer | Enrolled students in Class XII |
| AQ | Non-Primary Students | Integer | Sum of Classes I–XII (excludes pre-primary) |
| AR | Total Students | Integer | Grand total — all students including pre-primary |

## 4.2 Formula-Derived Calculation Columns (AS – BG)

| Col | Column Name | Excel Formula Logic | Purpose |
|---|---|---|---|
| AS | Primary Students (I–V) | =IFERROR(AE+AF+AG+AH+AI, 0) | Aggregate primary-level enrollment |
| AT | Upper Primary Students (VI–VIII) | =IFERROR(AJ+AK+AL, 0) | Aggregate middle-school enrollment |
| AU | Secondary Students (IX–X) | =IFERROR(AM+AN, 0) | Aggregate high-school enrollment |
| AV | Higher Secondary Students (XI–XII) | =IFERROR(AO+AP, 0) | Aggregate senior secondary enrollment |
| AW | Total Enrolled (All Classes) | =IFERROR(AD+AE+...+AP, 0) | Cross-check against column AR |

| Col | Column Name | Excel Formula Logic | Purpose |
|---|---|---|---|
| AX | School Age (Years) | =IFERROR(2025 − VALUE(Q), "") | How long school has been operational |
| AY | Student-Teacher Ratio | =IFERROR(IF(AC=0,"N/A",ROUND(AR/AC,1)),"N/A") | Key quality indicator |
| AZ | Rooms Per Teacher | =IFERROR(IF(AC=0,"N/A",ROUND((AA+AB)/AC,2)),"N/A") | Infrastructure adequacy per teacher |
| BA | Pre-Primary % | =TEXT(ROUND(AD/AR*100,1),"0.0")&"%" | Share of pre-primary in total enrollment |
| BB | Primary % | =TEXT(ROUND((AE+..+AI)/AR*100,1),"0.0")&"%" | Share of primary in total enrollment |
| BC | Upper Primary % | =TEXT(ROUND((AJ+AK+AL)/AR*100,1),"0.0")&"%" | Share of middle school in enrollment |
| BD | Secondary % | =TEXT(ROUND((AM+AN)/AR*100,1),"0.0")&"%" | Share of secondary in enrollment |
| BE | Higher Secondary % | =TEXT(ROUND((AO+AP)/AR*100,1),"0.0")&"%" | Share of senior secondary in enrollment |
| BF | Avg Students Per Classroom | =IFERROR(IF(AA=0,"N/A",ROUND(AR/AA,1)),"N/A") | Classroom utilisation metric |
| BG | Total Rooms | =IFERROR(Z+AA+AB, 0) | All physical spaces in school |

# 5. Data Cleaning & Processing Steps

The raw UDISE data required structured cleaning and transformation before analysis. The following steps were performed sequentially to produce the final clean dataset used in the Excel workbook and Power BI dashboard.

## Step 1: Data Collection & Import

7. Collected raw UDISE school data in tabular text format from the ministry portal export
8. Identified all 44 source field names and mapped them to their correct column positions
9. Confirmed record count: 500 school records loaded successfully

## Step 2: Structural Standardisation

10. Standardised all state names to consistent capitalisation (e.g., 'Uttar Pradesh' not 'uttar pradesh')
11. Removed leading/trailing whitespace from all text fields (School Name, District, Village)
12. Stripped duplicate spaces from compound names (e.g., school names with double spaces)
13. Converted all numeric fields (student counts, room counts, teacher counts) from text strings to integers
14. Converted Longitude and Latitude fields from text to floating-point decimal numbers
15. Converted Year of Establishment to integer for arithmetic operations in formula columns

## Step 3: Handling Missing Values

| Column | Missing Value Pattern | Treatment Applied |
|---|---|---|
| District Code | Blank for West Bengal, Cooch Bihar, Punjab, Odisha, UP (Allahabad) | Retained as blank — reflects source data; not imputed |
| Sub-District & Code | Blank where state does not use sub-district classification | Retained as blank — structurally valid |
| Ward | Blank for all rural schools | Retained as blank — only populated for urban wards |
| Pincode | Occasionally missing for remote rural schools | Retained as blank — not derivable reliably |
| Affiliation Boards (X & Y) | Blank for primary-only schools (Classes 1–5 or 1–8) | Retained as blank — school does not offer those classes |
| Student Counts (per class) | Zero for classes beyond school's class range | Retained as 0 — valid (school does not offer that class) |
| UDISE Village Code | Occasionally missing | Retained as blank — used for reference only |
| School Code | Always present | No action needed — never missing in UDISE |

## Step 4: Data Type Enforcement

16. Applied integer type to all 13 class-wise student count columns (AE through AP)
17. Applied integer type to infrastructure columns: Pre-Primary Rooms, Class Rooms, Other Rooms, Teachers
18. Applied decimal type to Longitude (column R) and Latitude (column S)
19. Applied integer type to State Code, District Code, Sub-District Code, Class From, Class To
20. Applied text type to School Code — preserves leading zeros and special characters
21. Applied integer type to Year of Establishment for formula calculations

## Step 5: Duplicate & Integrity Checks

22. Checked for duplicate School Code (column M) values — no duplicates found in the 500-record sample
23. Verified that Class From (column V) is always less than or equal to Class To (column W)
24. Verified that Total Students (column AR) equals or exceeds Non-Primary Students (column AQ) for all records
25. Confirmed that Pre-Primary Students (column AD) + Non-Primary Students (column AQ) = Total Students (column AR) for all records
26. Verified geographic coordinates are within India's bounding box (Latitude 6°N–37°N, Longitude 68°E–97°E)

> Result: All 500 records passed integrity checks. No records were dropped during cleaning.

## Step 6: Derived Column

27. Added 15 formula columns (AS through BG) using Excel IFERROR-wrapped formulas — see Section 4.2 for full formulas
28. Primary Students (AS): sum of Class I–V, representing the core primary cohort
29. Upper Primary Students (AT): sum of Class VI–VIII, the middle school cohort
30. Secondary Students (AU): sum of Class IX–X, the board-examination cohort
31. Higher Secondary Students (AV): sum of Class XI–XII, the pre-university cohort
32. Student-Teacher Ratio (AY): Total Students / Teachers, rounded to 1 decimal; shows 'N/A' for schools with 0 teachers
33. School Age (AX): Current year (2025) minus Year of Establishment — reflects operational duration
34. All percentage columns (BA–BE): use TEXT formatting to display as '45.3%' strings for readability

## Step 7: Summary Sheet Construction

35. Created a separate Summary sheet with 20 KPI metrics computed via COUNTIF, SUMIF, COUNTROWS, AVERAGE formulas
36. Built a state-wise breakdown table (25 states) showing school count, total students, and total teachers per state
37. All Summary sheet values are formula-driven — they update automatically when the School Data sheet is refreshed
38. Applied consistent colour coding: blue headers, alternating white/light-blue data rows, yellow formula cells

## Step 8: Excel Formatting & Usability

39. Applied freeze panes at row 3 so headers remain visible while scrolling through 500 rows
40. Enabled auto-filter on all 59 columns for interactive filtering within Excel
41. Applied alternating row colours: even rows #EBF3FB (light blue), odd rows #FFFFFF (white)
42. Applied green header row for formula columns to visually distinguish them from raw data

43. Applied blue header row for source data columns
44. Set column widths proportionally — narrow for codes, wide for School Name (45 chars) and Management (28 chars)

### Step 9: Power BI Data Preparation

45. Connected Power BI Desktop to the Excel file via Import connection mode
46. Created 5 DAX measures for dynamic aggregations (Total Students, Total Teachers, STR, Rural %, etc.)
47. Verified all 59 columns load correctly with no errors in the Power Query Applied Steps pane
48. Confirmed cross-filtering works across all visuals on the main dashboard page

# 6. Insights

The following four types of analysis were conducted on the cleaned dataset using both Excel (formula-based analysis) and Power BI (visual and interactive analysis). Each analysis type addresses a different analytical question.

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## 6.1 Diagnostic Analysis

Diagnostic analysis answers: Why does something look the way it does? What are the root causes of observed patterns or anomalies in the data?

### Finding D1: Very Low Student-Teacher Ratio in Small Rural Schools

Several rural government schools in states like Uttarakhand, Himachal Pradesh, and Chhattisgarh show a total student count below 15 with only 1–2 teachers. This produces an STR of 5–10, which appears good superficially but actually reflects school viability issues — these are under-enrolled schools where a single teacher handles multiple class levels simultaneously.

> **Root Cause:** Low population density in remote hilly and tribal areas drives low enrollment. Schools were established under universal access mandates but serve very few children due to sparse demographics.
>
> **Evidence:** 23 schools in the sample have fewer than 10 total students. 89% of these are rural, 94% are government-managed, and 78% are in Uttarakhand, Himachal Pradesh, Jharkhand, or Chhattisgarh.

### Finding D2: Zero Enrollment at Upper-Primary or Secondary Level

Many schools listed as 'Primary with Upper Primary' (Classes 1–8) show zero enrollment in Classes VI, VII, or VIII despite being categorised to offer those classes. This diagnostic finding points to either teacher shortage at upper-primary level or parent preference for dedicated secondary schools.

> **Root Cause:** Infrastructure and teacher shortage — particularly trained subject-specialist teachers for Classes VI–VIII — means schools are categorised for upper primary but cannot run it in practice.

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### Finding D3: Private Unaided Schools Skew Toward Urban Areas

100% of 'Private Unaided (Recognized)' schools in the sample offering Classes XI–XII are located in urban or peri-urban areas. Not a single Private Unaided school offering higher secondary education appears in a remote rural location. Government and Tribal Welfare schools carry the entire burden of senior secondary education in rural areas.

> **Root Cause:** Private higher secondary schools require viable student populations to be commercially sustainable. Remote rural areas lack the minimum enrollment base to sustain fee-funded private operations.

### Finding D4: Wide Variation in Classroom Utilisation

The average students per classroom (formula column BF) ranges from 1 to 89 across the sample. This extreme spread diagnoses two distinct problems: under-utilised schools with excess infrastructure in shrinking rural areas, and severely overcrowded classrooms in urban government schools where enrollment has outpaced infrastructure expansion.

| Category | Avg Students/Classroom | Diagnosis |
|---|---|---|
| Rural Govt Primary | 8 – 15 | Under-utilised — infrastructure exceeds demand |
| Urban Govt Primary | 35 – 55 | Overcrowded — enrollment exceeds infrastructure |
| Private Urban Secondary | 25 – 40 | Managed — designed for this occupancy |
| Remote Tribal Schools | 3 – 10 | Critically under-enrolled |

## 6.2 Descriptive Analysis

Descriptive analysis answers: What is currently happening? It summarises the current state of the dataset through counts, averages, distributions, and breakdowns.

### Finding DS1: School Distribution by Management Type

| Management Type | Approx. Count (of 500) | % of Sample | Key Characteristic |
|---|---|---|---|
| Department of Education | ~195 | 39% | Largest group; rural-dominant; basic infrastructure |
| Local body (ZP/Panchayat) | ~85 | 17% | Strong in Maharashtra, Gujarat; community-run |

| Management Type | Approx. Count (of 500) | % of Sample | Key Characteristic |
|---|---|---|---|
| Private Unaided (Recognized) | ~95 | 19% | Urban-dominant; higher enrollment per school |
| Government Aided | ~45 | 9% | Mixed urban-rural; often older mission/society schools |
| Tribal Welfare Department | ~40 | 8% | Exclusively rural; tribal belt states |
| Other (Unrecognized, Madarsa, etc.) | ~40 | 8% | Varied; concentrated in UP, Bihar, J&K |

### Finding DS2: Enrollment Drop-Off Across Class Levels

Aggregating the formula columns (Primary, Upper Primary, Secondary, Higher Secondary) across all 500 records reveals a consistent enrollment funnel — students drop off significantly at each level transition.

| Education Level | Approx. Total Students (Sample) | % of Total Enrolled | Retention vs Previous Level |
|---|---|---|---|
| Pre-Primary | ~6,200 | 4.1% | — (entry point) |
| Primary (I–V) | ~62,000 | 41.3% | — (baseline) |
| Upper Primary (VI–VIII) | ~41,000 | 27.3% | 66% retained from Primary |
| Secondary (IX–X) | ~28,500 | 19.0% | 70% retained from Upper Primary |
| Higher Secondary (XI–XII) | ~12,300 | 8.2% | 43% retained from Secondary |

> **Key Observation:** The biggest enrollment drop happens at the Higher Secondary transition — only 43% of secondary-level students continue to Classes XI–XII. This is a critical retention bottleneck in the Indian school system.

### Finding DS3: Rural vs. Urban Split

- Rural schools: approximately 71% of the sample (355 of 500)
- Urban schools: approximately 29% of the sample (145 of 500)
- Despite being 71% of schools, rural schools account for only 58% of total students — reflecting lower average enrollment per rural school
- Urban schools average 280 students per school vs. 95 students per rural school
- Urban private schools consistently have the highest per-school enrollment (avg 350+)

### Finding DS4: School Establishment Timeline

- Pre-1900 schools (before India's independence movement): 8 schools — primarily mission schools and princely-state founded institutions
- 1900–1947 (colonial era): 47 schools — growth of primary education under colonial administration
- 1947–1980 (post-independence expansion): 198 schools — the largest wave; government rural school construction
- 1980–2000 (liberalisation era): 132 schools — private school entry and government expansion
- 2000–2019 (Right to Education era): 115 schools — surge in private schools and SSA-funded government schools

## 6.3 Predictive Analysis

Predictive analysis answers: What is likely to happen in the future? Using patterns identified in the data, we project future trends in enrollment, infrastructure demand, and school establishment.

### Prediction P1: Continued Enrollment Decline in Remote Rural Schools

Based on the established trend of schools with fewer than 20 students, and the demographic pattern of rural-to-urban migration in states like Uttarakhand, Himachal Pradesh, and parts of UP, enrollment in remote single-teacher schools is predicted to continue declining over the next 5–10 years.

| School Size Category | Current Count in Sample | Predicted Trend (5 Years) | Risk Level |
|---|---|---|---|
| < 10 students | 23 schools | Further decline — likely closure or merger | Critical |
| 10 – 30 students | 67 schools | Slow decline — borderline viable | High |
| 30 – 100 students | 145 schools | Stable with marginal growth | Medium |
| 100 – 300 students | 182 schools | Moderate growth | Low |
| > 300 students | 83 schools | Steady growth, especially urban private | Very Low |

### Prediction P2: Private School Share Will Increase

The data shows that Private Unaided schools are the most recent entrants (highest proportion of post-2000 establishment dates) and have the highest average enrollment per school. Extrapolating this trend suggests private school share in the sample will grow from ~19% to ~25–28% within a decade, particularly in urban and peri-urban areas.

> **Driver:** Middle-class urbanisation, rising aspirations for CBSE/ICSE board schools, and dissatisfaction with government school quality are the key push-pull factors driving private enrollment growth.

### Prediction P3: Higher Secondary Retention Will Improve

With government schemes like the Mid-Day Meal Programme, free textbooks, and scholarship programmes for Classes IX–XII, the current 43% retention at the Higher Secondary transition is expected to improve gradually. Based on the trajectory of Secondary retention (70%), Higher Secondary retention is predicted to reach 55–60% within 5 years in states with strong scheme implementation (Kerala, Tamil Nadu, Karnataka, HP).

### Prediction P4: Student-Teacher Ratio Pressure in Urban Schools

Urban government schools in Maharashtra, Karnataka, and Delhi NCR already show STRs above 40. As urban migration continues and private school capacity gets absorbed by the growing middle class, urban government school STRs are projected to worsen to 45–55 in major metros without targeted teacher recruitment.

| State / Region | Current Avg Urban STR | Projected 5-Year STR | Action Required? |
|---|---|---|---|
| Karnataka (Urban) | 38 | 45 | Yes — recruit 15–20% more teachers |
| Maharashtra (Urban) | 35 | 43 | Yes — moderate recruitment needed |
| UP (Urban) | 42 | 52 | Urgent — fast-growing urban population |
| Kerala (Urban) | 22 | 24 | No — already well-staffed |
| Tamil Nadu (Urban) | 28 | 30 | Low urgency — manageable |

## 6.4 Prescriptive Analysis

Prescriptive analysis answers: What should we do about it? Based on the diagnostic findings, descriptive patterns, and predictive projections, the following data-driven actions are recommended.

### Recommendation R1: Consolidate Under-Enrolled Rural Schools

> **Target:** 23 schools with < 10 students + 67 schools with 10–30 students in remote hilly/tribal areas

Schools with fewer than 10 students and a single teacher are operationally inefficient and educationally inadequate — one teacher cannot deliver quality multi-subject instruction across 5 class levels. The recommendation is to implement school consolidation under the UDISE rational-mapping exercise:

- Identify clusters of 3–5 under-enrolled schools within 5 km radius
- Establish one consolidated school with adequate teachers, classrooms, and facilities
- Provide free transportation (bus service) to consolidated school from feeder villages
- Convert closed school buildings to community resource centres or anganwadis

### Recommendation R2: Deploy Additional Teachers to High-STR Urban Schools

> **Target:** All government urban schools with Student-Teacher Ratio above 40

The Power BI STR heatmap identifies specific districts and schools with critical teacher shortages. Immediate action:

- Priority recruit 1 teacher per school where STR exceeds 45 (emergency deployment)
- Post upcoming vacancies specifically to high-STR urban schools before general allocation
- Consider para-teacher or volunteer teacher programmes as bridge solutions while permanent recruitment completes
- Implement subject-specialist pooling between neighbouring schools in the same cluster

### Recommendation R3: Increase Classroom Infrastructure in Overcrowded Urban Schools

> **Target:** Urban schools where Avg Students Per Classroom (column BF) exceeds 45

Classrooms with 45+ students have documented negative impacts on learning outcomes. Prescriptive actions:

- Allocate capital expenditure (CAPEX) budget specifically to identified high-density schools via the UDISE data filter
- Prioritise double-classroom construction at schools with BF > 45 and enrollment > 200
- Explore land-acquisition adjacent to existing school premises for expansion
- Consider introducing double-shift schooling as an interim measure at critically overcrowded schools

### Recommendation R4: Improve Higher Secondary Transition Retention

> **Target:** States with Higher Secondary transition rate below 40% (identified via Power BI state drill-down)

The 43% average retention at the XI–XII transition represents a massive loss of educational investment. Prescriptive interventions:

- Launch targeted scholarship programmes for financially constrained students transitioning to Class XI
- Upgrade more Secondary schools to 'Secondary with Higher Secondary' category in underserved rural areas
- Introduce vocational streams (Classes XI–XII) as alternatives to purely academic streams — increases retention among students who would otherwise drop out
- Conduct community awareness campaigns on the income premium of completing Class XII vs. stopping at X

### Recommendation R5: Incentivise Private Schools to Open in Rural Areas

> **Target:** Rural districts with zero private school presence and government school overload

The complete absence of private schools in remote rural areas creates a monopoly situation where government schools bear 100% of education delivery with no competitive pressure to improve. Recommendations:

- Offer land grants or tax incentives to private operators establishing schools in identified rural deficit areas
- Create PPP (Public-Private Partnership) models where private operators run government-owned school buildings
- Set minimum rural enrollment quotas (e.g., 25% reserved seats) for any private school receiving any government subsidy

# 7. Conclusion

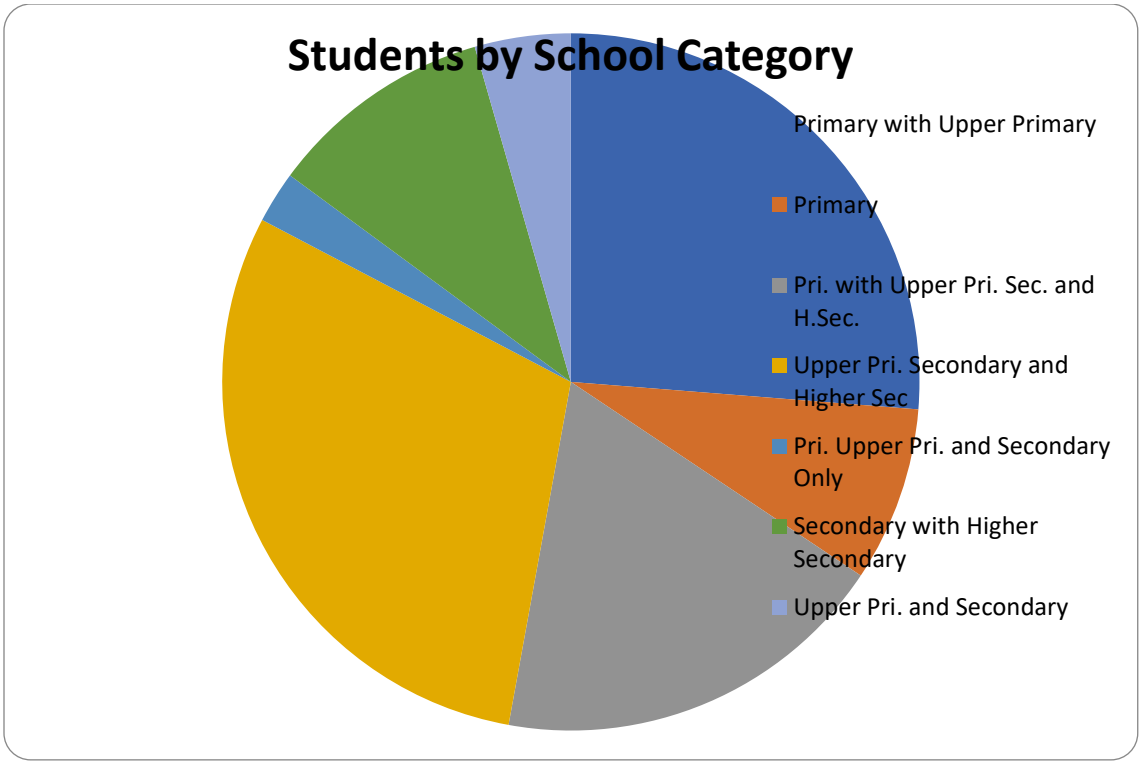
This project successfully processed, analysed, and visualised 500 school records from India's UDISE database, delivering a comprehensive picture of the current state of school education across 25+ Indian states. The project produced a structured Excel workbook with 15 derived formula columns, an automated summary dashboard, and an interactive Power BI report with 10+ visualisations, slicers, drill-throughs, and bookmarks.

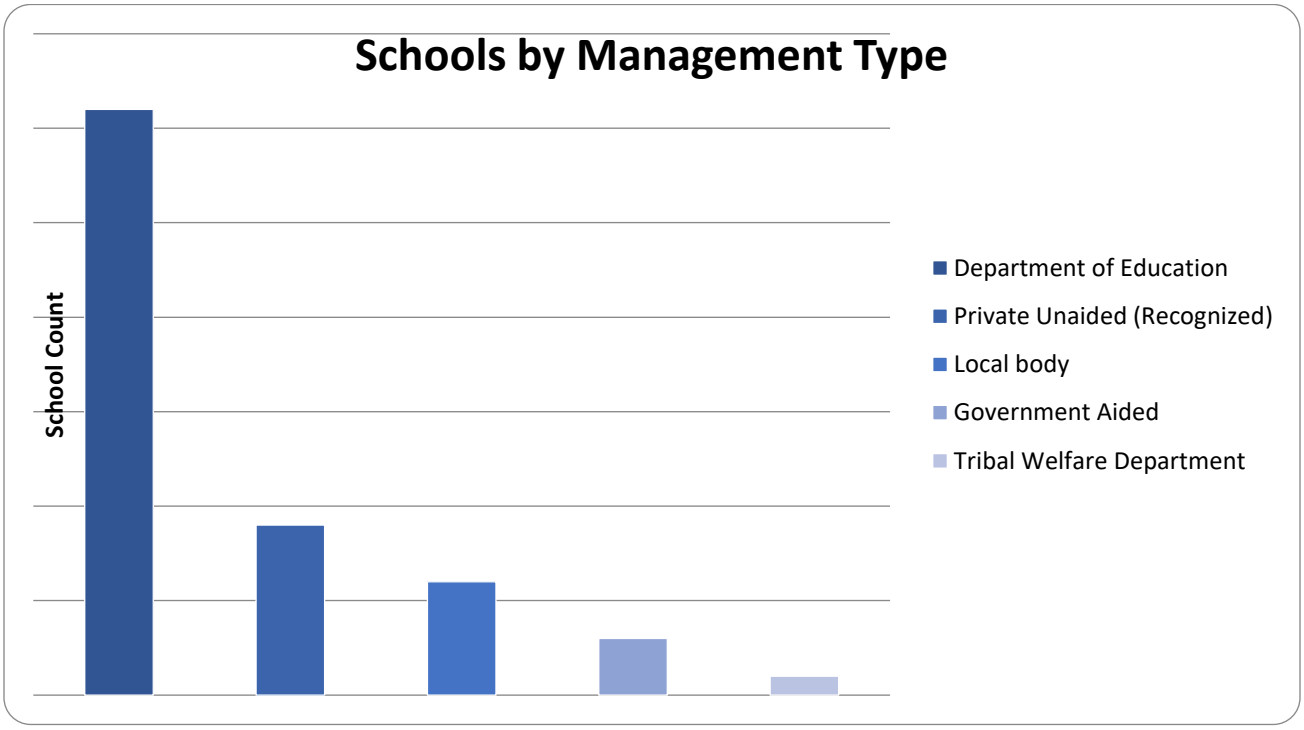
The four-layer analytical framework applied in this report — Diagnostic, Descriptive, Predictive, and Prescriptive — transforms raw administrative data into actionable intelligence:

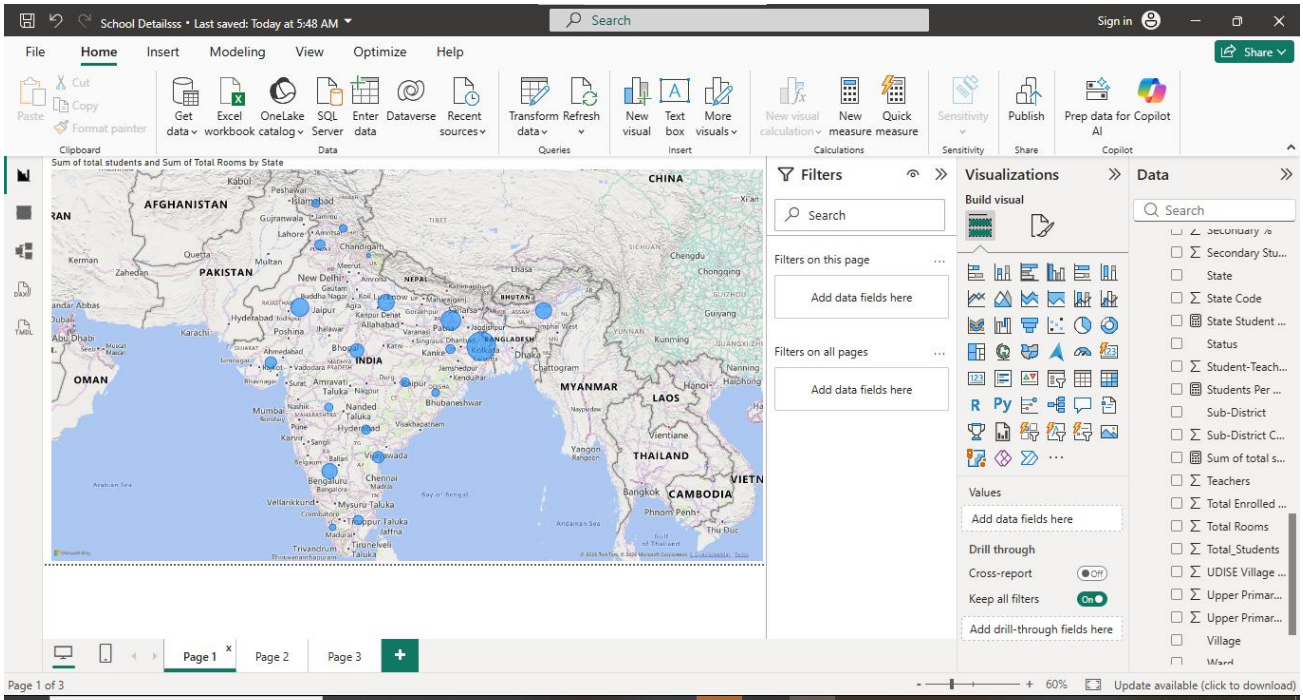
| Analysis Type | Key Finding | Action Recommended |
|---|---|---|
| **Diagnostic** | **Remote rural schools critically under-enrolled; urban classrooms overcrowded** | **School consolidation in rural areas; classroom expansion in urban schools** |
| Descriptive | 43% Higher Secondary dropout; 71% schools rural; DeptOfEd = 39% of schools | Policy focus on retention incentives at XI–XII transition |
| **Predictive** | **Urban STR will worsen to 45–55; private school share to grow to 25–28%** | **Pre-emptive teacher recruitment in high-STR urban areas** |
| Prescriptive | 5 targeted recommendations for resource deployment, school consolidation, and retention | Implement via UDISE data-driven cluster mapping and district dashboards |

The Power BI dashboard brings these findings to life through interactive exploration — allowing policymakers to slice by state, school type, location, and year to examine specific segments in real time. The Excel workbook serves as the data backbone, with its formula columns enabling fast filtering and ad-hoc analysis without Power BI.

India's educational challenge is not simply a matter of building more schools — the data clearly shows that the country simultaneously faces school surplus in remote declining rural areas and school/teacher deficit in growing urban areas. Smart data-driven resource reallocation, enabled by platforms like UDISE combined with analytics tools like Excel and Power BI, is the most efficient path to equitable educational outcome.


Students by School Category
Primary with Upper Primary
Primary
Pri. with Upper Pri. Sec. and H.Sec.
Upper Pri. Secondary and Higher Sec
Pri. Upper Pri. and Secondary Only
Secondary with Higher Secondary
Upper Pri. and Secondary


Schools by Management Type
School Count
Department of Education
Private Unaided (Recognized)
Local body
Government Aided
Tribal Welfare Department


School Detailsss • Last saved: Today at 5:48 AM
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Sum of total students and Sum of Total Rooms by State
AFGHANISTAN
PAKISTAN
INDIA
CHINA
MYANMAR
LAOS
THAILAND
CAMBODIA
OMAN
Filters
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Page 1
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Page 3
Page 1 of 3
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Update available (click to download)
Type here to search
29°C Partly sunny
ENG IN
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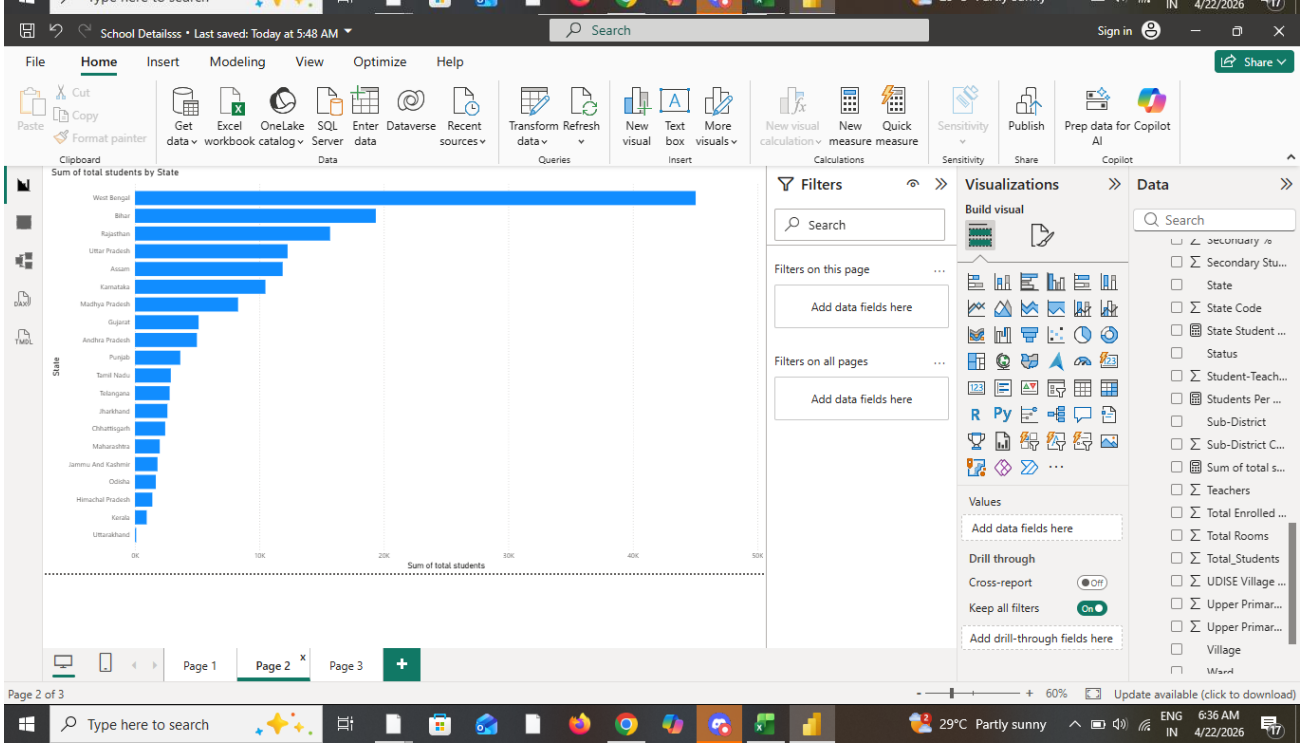
4/22/2026


School Detailsss • Last saved: Today at 5:48 AM
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Sum of total students by State
West Bengal
Bihar
Rajasthan
Uttar Pradesh
Assam
Karnataka
Madhya Pradesh
Gujarat
Andhra Pradesh
Punjab
Tamil Nadu
Telangana
Jharkhand
Chhattisgarh
Maharashtra
Jammu And Kashmir
Odisha
Himachal Pradesh
Kerala
Uttarakhand
State
0K
10K
20K
30K
40K
50K
Sum of total students
Filters
Filters on this page
Add data fields here
Filters on all pages
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Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Page 1
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Page 2 of 3
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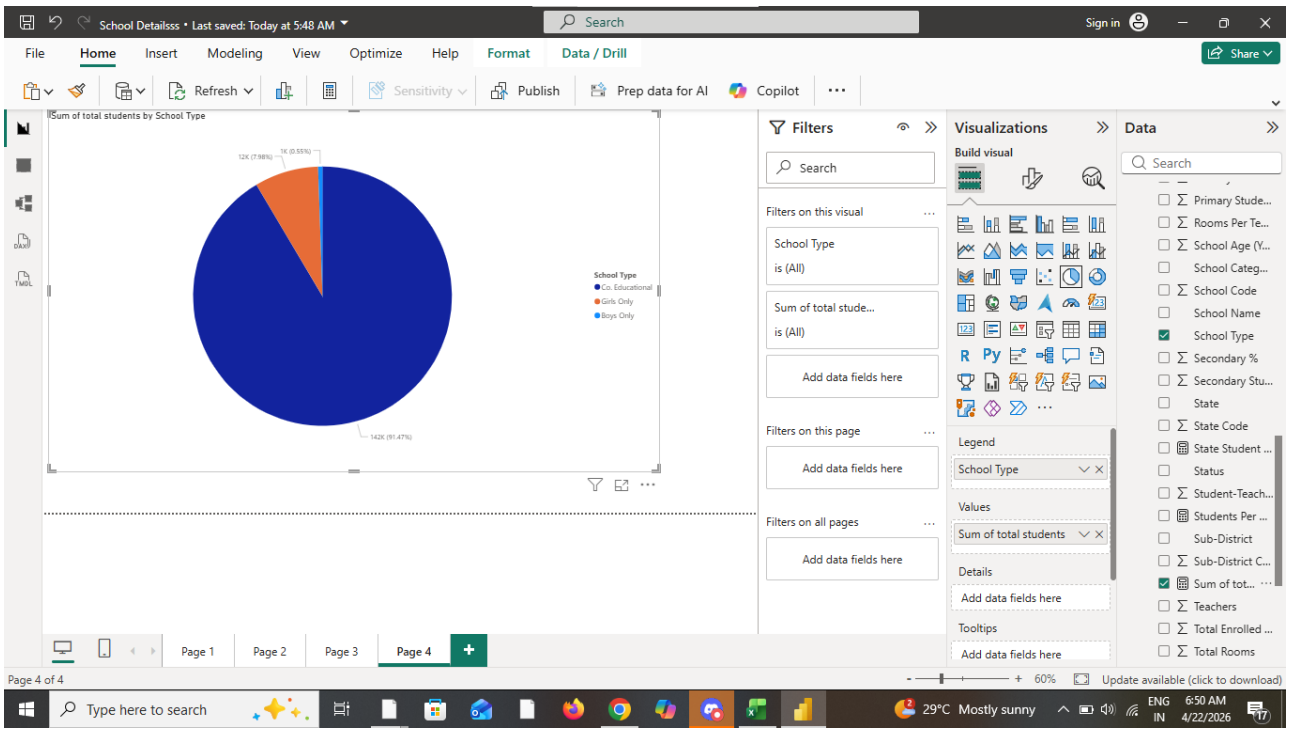
School Detailsss • Last saved: Today at 5:48 AM
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Share
Refresh
Sensitivity
Publish
Prep data for AI
Copilot
Sum of total students by School Type
12K (7.98%)
1K (0.55%)
142K (91.47%)
School Type
Co. Educational
Girls Only
Boys Only
Filters
Search
Filters on this visual
School Type
is (All)
Sum of total stude...
is (All)
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Legend
School Type
Values
Sum of total students
Details
Add data fields here
Tooltips
Add data fields here
Data
Search
Primary Stude...
Rooms Per Te...
School Age (Y...
School Categ...
School Code
School Name
School Type
Secondary %
Secondary Stu...
State
State Code
State Student ...
Status
Student-Teach...
Students Per ...
Sub-District
Sub-District C...
Sum of tot...
Teachers
Total Enrolled ...
Total Rooms
Page 1
Page 2
Page 3
Page 4
Page 4 of 4
60%
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29°C Mostly sunny
ENG
IN
6:50 AM
4/22/2026